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RAJAGIRI SCHOOL OF
ENGINEERING & TECHNOLOGY
(AUTONOMOUS)

REPORT

Submitted by

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INEFFICIENT CAMPUS RESOURCE MANAGEMENT

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Abstract

This technical report addresses Inefficient Campus Resource Management, a critical issue inflating the carbon footprint and operational costs of educational institutions. Campuses currently rely on static operational schedules, leading to massive energy waste in unoccupied lecture halls and unmonitored plumbing leaks that remain undetected between billing cycles. To resolve this, this report proposes a unique Engineering/IT-based solution: an IoT-based Predictive Resource Allocation System. Designed as the intelligent backend for the group's Green Campus Management Portal, this system utilizes distributed microcontrollers, passive infrared sensors, and digital flow meters to replace manual audits with automated utility actuation. A comprehensive sustainability analysis validates this solution against ISO 14001 and specific SDGs (7, 9, 12). By neutralizing human error, this scalable IoT architecture ensures rapid economic returns, curtails operational emissions, and guarantees environmental compliance.

Introduction

Contemporary higher education institutions operate on a scale akin to small municipalities, yet their facility management protocols often remain archaic. The resulting Inefficient Campus Resource Management severely undermines institutional commitments to sustainability. At the core of this inefficiency is a persistent reliance on rigid, time-based utility scheduling rather than dynamic, occupancy-based resource allocation.

Academic environments experience highly erratic utilization. High-capacity HVAC units and lighting operate on fixed timers, generating phantom electrical loads in completely empty rooms. Simultaneously, plumbing infrastructure lacks real-time sub-metering, allowing minor pipe fractures to leak thousands of liters of clean water daily before facility managers are alerted via inflated monthly bills.

Historically, institutions relied on behavioral awareness campaigns (e.g., "turn off the lights" posters), which consistently fail due to unreliable human compliance. This report delineates a paradigm shift toward proactive, automated conservation. By deploying a deterministic, sensor-driven IoT network, the proposed system actively monitors room occupancy and fluid distribution, triggering immediate physical actuations to isolate lighting circuits and shut off leaking valves. This real-time telemetry directly powers the interactive Green Campus Management Portal, uniting an automated backend with user-facing web technologies to establish a verifiable, zero-waste smart campus

Problem Statement

Inefficient Campus Resource Management results in excessive energy consumption, severe water wastage, and inflated operational costs. Without real-time monitoring infrastructure, institutions rely on reactive manual audits and static operational schedules. Consequently, academic buildings cool and illuminate empty lecture halls, generating massive phantom electrical loads. Simultaneously, a lack of sub-metering means undetected plumbing leaks waste thousands of liters of water before being caught during delayed monthly billing cycles.

Sustainability Analysis

- **Relevant SDGs:** This systemic inefficiency directly hinders SDG 7 (Affordable and Clean Energy) by artificially inflating base-load energy demand, SDG 9 (Industry, Innovation, and Infrastructure) through the continued reliance on legacy unmonitored systems, and SDG 12 (Responsible Consumption and Production) via the continuous over-extraction of municipal water resources.
- **ISO 14001 Integration:** Deploying an automated IoT monitoring system aligns seamlessly with ISO 14001 standards by establishing a definitive, data-driven framework for continuous environmental performance improvement and automated compliance tracking.
- **Life Cycle Assessment (LCA) / EIA:** An LCA of campus facilities indicates that the operational phase—specifically daily electricity and water consumption—contributes the vast majority of the institution’s overall carbon footprint. Targeting this operational phase with real-time utility mitigation provides the most effective intervention point to reduce Scope 2 greenhouse gas emissions

Proposed Engineering/IT Solution

My unique contribution to the group's Green Campus Management Portal is an IoT-based Predictive Resource Allocation System.

- **Hardware Layer:** Deployment of ESP32 microcontrollers, Passive Infrared (PIR) occupancy sensors, and Hall-effect digital water flow meters across campus zones.
- **Middleware:** MQTT protocol transmits telemetry to a cloud broker, where an algorithm cross-references motion triggers with academic scheduling APIs.
- **Actuation & Integration:** The system automatically triggers relay modules to isolate lighting circuits and motorized valves to shut off leaking water lines. This real-time data feeds directly into the responsive web portal (built with HTML, CSS, and JavaScript) for live monitoring.

Feasibility Analysis

- **Technical:** Highly viable using open-source hardware, standard web protocols, and RESTful APIs to integrate with the frontend portal.
- **Economic:** Capital expenditure for sensor nodes is rapidly offset by a projected 20-30% reduction in monthly utility overhead.
- **Operational:** The modular system operates invisibly, requiring zero behavioral modifications from the student body.

Expected Environmental Benefits

- **Energy Reduction:** Automating HVAC and lighting based on true occupancy eliminates phantom loads, reducing base-load electricity consumption by an estimated 25%.
- **Water Conservation:** Immediate actuation upon detecting anomalous continuous flow prevents the loss of thousands of liters of clean water annually.
- **Emission Mitigation:** Directly reduces Scope 2 greenhouse gas emissions associated with institutional operations.

Justification of the Proposed Solution

Unlike traditional awareness campaigns that rely on unreliable human compliance, this automated system enforces resource conservation deterministically. It neutralizes human error and guarantees environmental compliance, providing the exact real-time data required to power the interactive features of the group's web application.

Conclusion

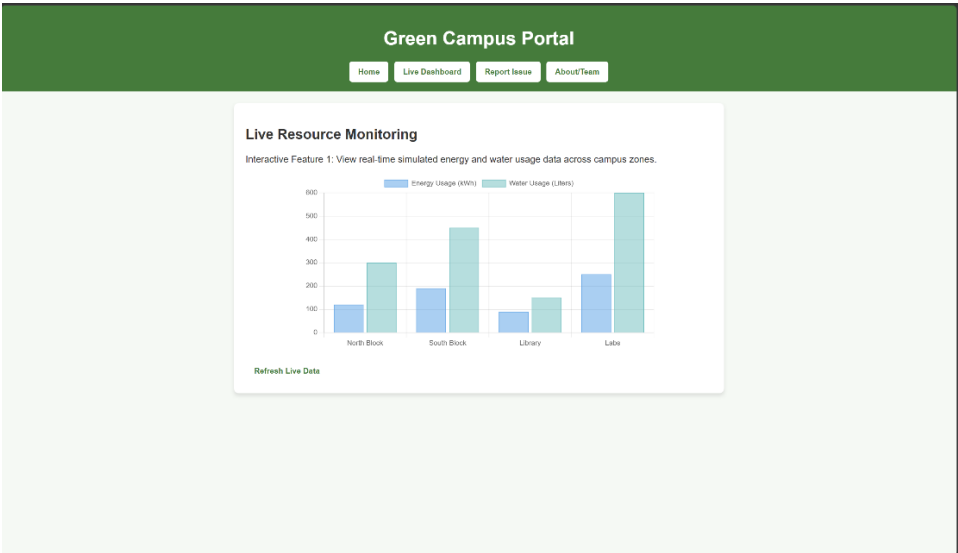
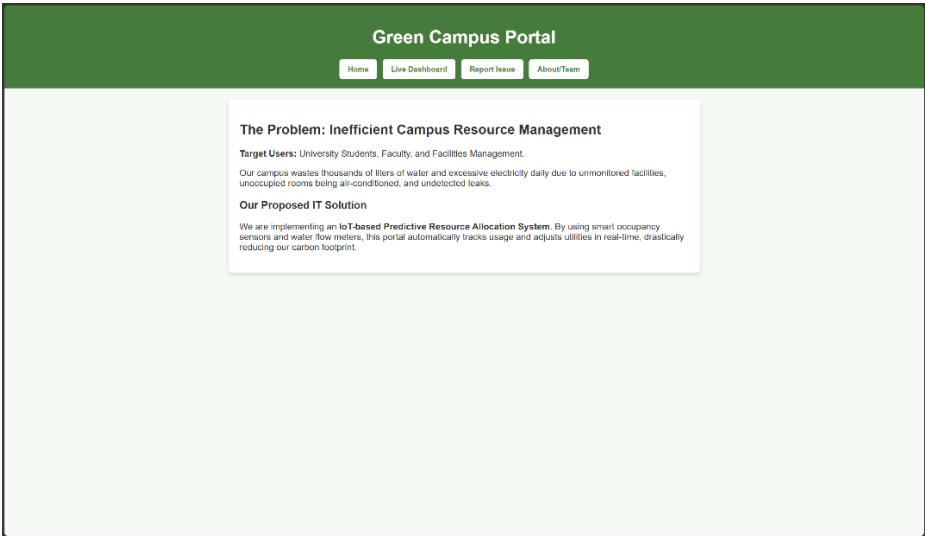
The pervasive issue of Inefficient Campus Resource Management requires a definitive shift from reactive facility maintenance to proactive, automated conservation. By engineering an IoT-based Predictive Resource Allocation System as the backend intelligence for the Green Campus Management Portal, this report demonstrates a viable, scalable solution to structural utility wastage. Integrating ESP32 microcontrollers, passive infrared sensors, and digital flow meters completely removes human negligence from the conservation equation. The resulting automated actuations immediately mitigate phantom electrical loads and undetected water leaks, sharply reducing the institution's operational carbon footprint. As validated through the lens of ISO 14001 and the UN Sustainable Development Goals (SDGs 7, 9, and 12), deploying this data-driven architecture ensures strict environmental compliance while providing the real-time telemetry required to operate a truly sustainable, smart campus.

References

(Format these according to your institution's required citation style, such as APA or IEEE. Ensure you add specific URLs or textbook pages you referenced during your study.)

1. United Nations. (2015). *Transforming our world: the 2030 Agenda for Sustainable Development* (SDG 7, 9, 12).
2. International Organization for Standardization. (2015). *ISO 14001:2015 Environmental management systems — Requirements with guidance for use*.
3. Espressif Systems. (2023). *ESP32 Series Datasheet*. (Used for technical feasibility of the microcontroller network).
4. Course Material: CO900G - *Environmental Science and Sustainable Engineering*, Alternative Assessment Method Guidelines.
5. Mozilla Developer Network (MDN). (2023). *HTML, CSS, and JavaScript Web Documentation*. (Referenced for frontend portal development).

Appendix: Website Screenshots



Green Campus Portal

Home Live Dashboard Report Issue About Team

Report an Environmental Issue

Interactive Feature 2: Dynamic form that calculates estimated resource loss based on your report.

Type of Issue:

Select an option...

Estimated hours active:

4

Calculate Impact & Submit

⚠️ Estimated Loss: 75 liters of water wasted! Issue reported to maintenance.